



Serial No: PE2363P6881
Drawing No:1.710.3473

1 Description			
Name of Product	Three-phase Pad Mount Transformer		
Serial No.	PE2363P6881		
Drawing No.	1.710.3473		
2 Technical Specification			
Power Rating	1500kVA	Cooling Type	ONAN
Rated Voltage	12.47/0.48kV	Number of Phases	3 Phases
Rated Current	69.45/1804.2A	Rated Frequency	60 Hz
Vector Group	YNyn0		
Basic Insulation Level	HV	LI/AC 95/34 kV	
	LV	LI/AC 30/10 kV	
3 Applicable standards			
ANSI/IEEE C57.12.00 General Requirements for Liquid-Immersed Distribution, Power and Regulating Transformers ANSI/IEEE C57.12.90 Test Code for Liquid-Immersed Distribution, Power and Regulating Transformers ANSI/IEEE C57.12.80 Standard Terminology for Power and Distribution Transformers ANSI/IEEE C57.100 IEEE Standard Test Procedure for Thermal Evaluation of Liquid-Immersed Distribution and Power Transformers ANSI/IEEE C57.136 IEEE Guide For Sound Level Abatement & Determination For Liquid-Immersed Power Transformers & Shunt Reactors Rated Over 500 KVA ASTM D 6871 Standard Specification for Natural (Vegetable Oil) Ester Fluids Used in Electrical Apparatus AWS American Welding Society NEMA National Electrical Manufacturers Association NFPA TO National Fire Protection Association National Electrical Code SSPC The Society for Protective Coatings Paint Application Specification NO. 2			

4 Items and results of tests

4.1 Measurement of voltage ratio and check of vector group

H.V.		L.V.		Calculated Ratio	Error (%)						Tolerance	Vector Group
Tap Position	Voltage (V)	Tap Position	Voltage (V)		H1H2/X1X2		H2H3/X2X3		H3H1/X3X1			
1	13094	/	480	27.279	0.00	0.01	0.00	0.01	0.00	0.02	≤±0.5%	YNyn0
2	12782			26.629	0.09	0.11	0.09	0.10	0.09	0.10		
3	12470			25.979	0.18	0.20	0.18	0.20	0.18	0.20		
4	12158			25.329	0.04	0.05	0.04	0.05	0.04	0.05		
5	11847			24.681	0.14	0.14	0.14	0.14	0.14	0.14		

Measuring instrument: Type: JYT(A)

4.2 Measurement of winding resistance				
Average Oil Temperature:14.1℃				
Tap Position	H.V. Windings (Ω)			Unbalance Rate
	H1H2	H2H3	H3H1	Measured
1	0.7589	0.7612	0.7599	0.30%
2	0.7399	0.7419	0.7409	0.27%
3	0.7205	0.7222	0.7215	0.24%
4	0.6998	0.7012	0.7009	0.20%
5	0.6807	0.6818	0.6818	0.16%
Tap Position	L.V. Windings (Ω)			Unbalance Rate
	X1X2	X2X3	X3X1	Measured
/	0.0008996	0.0008969	0.0009040	0.79%
	X1X0	X2X0	X3X0	Measured
/	0.0004921	0.0004855	0.0004826	1.95%
Measuring instrument: Type JYR(10C)				

4.3 Measurement of Insulation Resistance								
Relative Humidity:71%,Top Oil Temperature:14.1℃								
Measured Between	1min Insulation Resistance (MΩ)		Voltage (V)					
H.V.~L.V. &Earth	> 2500		2500					
L.V.~H.V.& Earth	> 5000							
H.V. & L.V.~Earth	> 2500							
Measuring Instrument: 3125A.								
4.4 Applied Voltage Test (AV)								
Measured Between	Test Voltage (kV)	Test Frequency (Hz)	Duration (s)	Result				
H.V.	34	50	60	pass				
L.V.	10	50	60	pass				
4.5 Induced Voltage Withstand Test(IVW)								
Tap Position	Tested Winding	Supplied Terminal	Supplied Voltage (kV)	Induced Voltage (kV)	Induced Multiple	Test Frequency (Hz)	Duration (S)	Result
3	H.V.	L.V.	0.96	24.94	2	150	40	Pass

4.6 Measurement of No-load Current and No-load Loss

Average Oil Temperature: 14.1 °C

Test Voltage (kV)		Test Current			No-load Loss (kW)		
Mean	R.M.S	I0(A)	I0(%)		Test Loss (Pm)	Corrected Loss (P0)	Required
			Required	Measured			
0.48	0.50	5.59	/	0.26	1.095	1.456	<1.4+10%

Measuring Instrument: Norma 4000CN precision power analyzer.

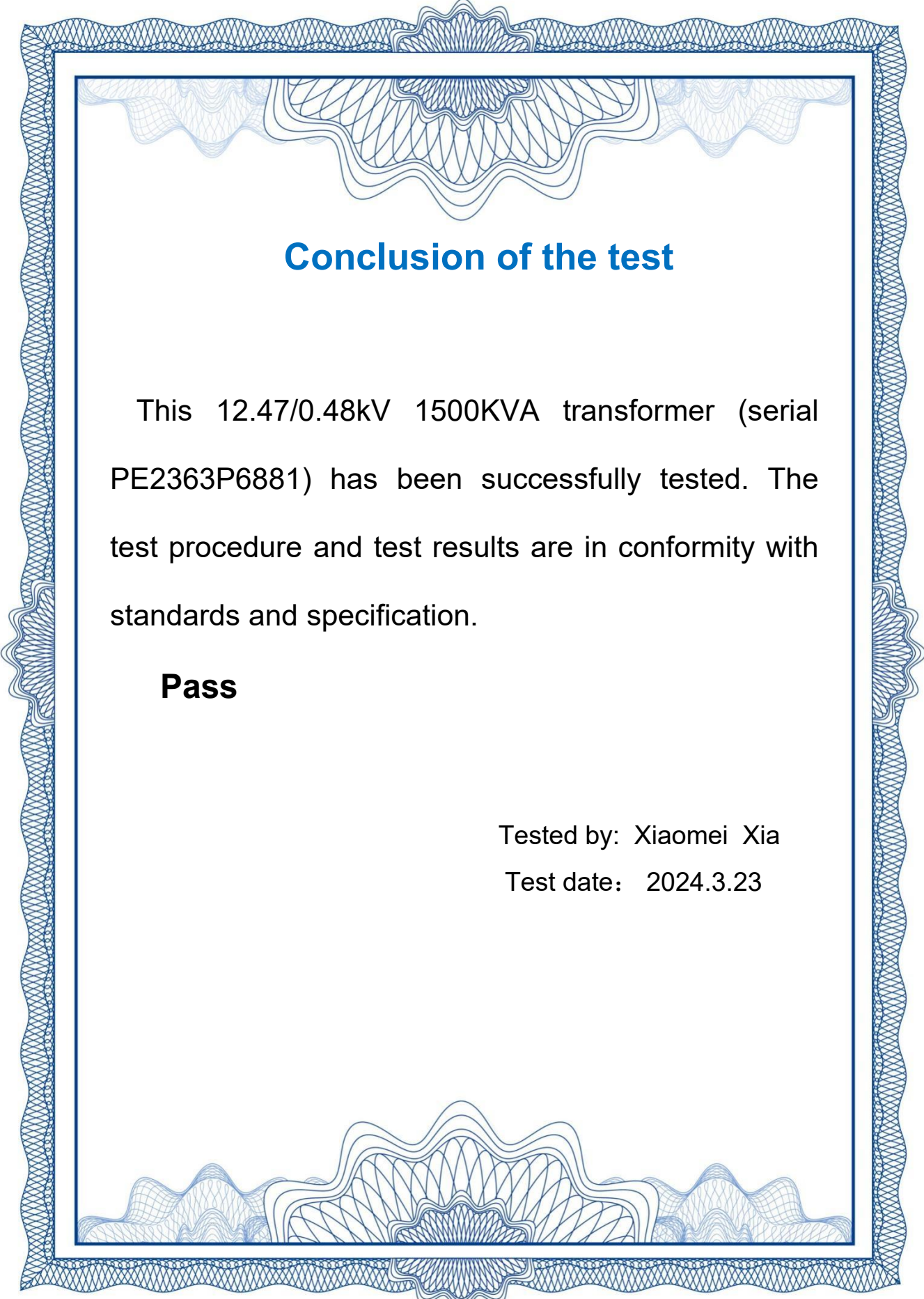
4.7 Measurement of Short-circuit Impedance and Load Loss

【H.V.~L.V.】

Tap Position		Test Current I(A)	Test Voltage U(kV)	Impedance Base:1500kVA				Temp.:75°C(kW) Base:1500kVA	Temp.:85°C(kW) Base:1500kVA	
H.V.	L.V.			Required	Tolerance	Uk(%)	Zk(Ω)	Load Loss	Required	Load Loss
3	/	41.17	0.389	/	/	6.35	/	13.327	<14+6%	13.767

Measuring Instrument: Norma 4000CN precision power analyzer.

4.8 Tank Pressure (Leakage) Test				
Test Method	Applied Pressure (MPa)	Duration of Test (h)	Release Pressure (Mpa)	Result
Tank Pressure	0.03	24	0.03	No leaks



Conclusion of the test

This 12.47/0.48kV 1500KVA transformer (serial PE2363P6881) has been successfully tested. The test procedure and test results are in conformity with standards and specification.

Pass

Tested by: Xiaomei Xia

Test date: 2024.3.23